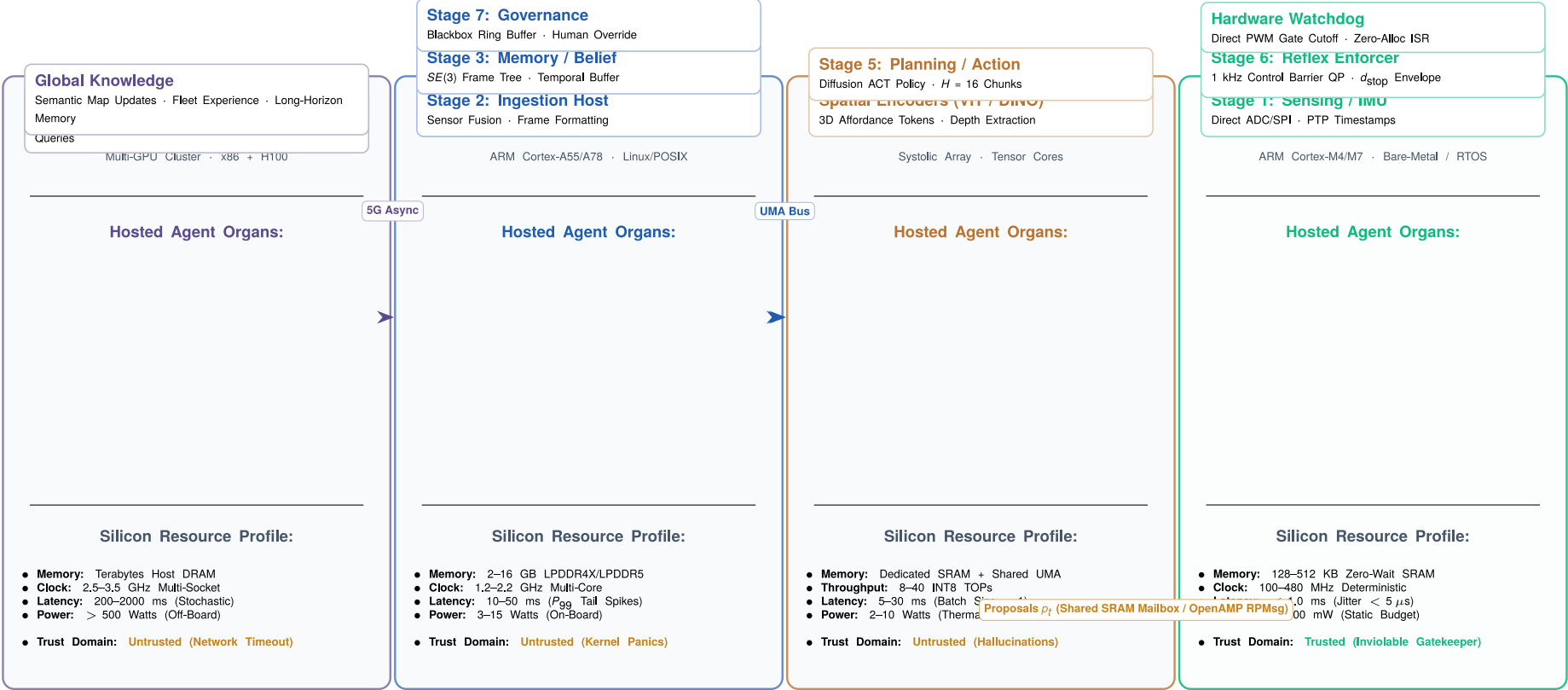




Stage 5: Planning / Action

Diffusion ACT Policy · H = 16 Chunks

Spatial Encoders (VIT / DINO)

3D Affordance Tokens · Depth Extraction

Systolic Array · Tensor Cores

Hosted Agent Organs:

Silicon Resource Profile:

- Memory: Dedicated SRAM + Shared UMA
- Throughput: 8–40 INT8 TOPs
- Latency: 5–30 ms (Batch Scheduling)
- Power: 2–10 Watts (Thermal)

Trust Domain: Untrusted (Hallucinations)

Hardware Watchdog

Direct PWM Gate Cutoff · Zero-Alloc ISR

Stage 6: Reflex Enforcer

1 kHz Control Barrier QP · d_{stop} Envelope

Stage 1: Sensing / IMU

Direct ADC/SPI · PTP Timestamps

ARM Cortex-M4/M7 · Bare-Metal / RTOS

Hosted Agent Organs:

Silicon Resource Profile:

- Memory: 128–512 KB Zero-Wait SRAM
- Clock: 100–480 MHz Deterministic
- Latency: 10–100 μs (Jitter < 5 μs)
- Power: 100 mW (Static Budget)

Trust Domain: Trusted (Inviolable Gatekeeper)

INTERCONNECT BUS TOPOLOGY & QUALITY-OF-SERVICE (QoS) ARBITRATION:

5G / Wi-Fi 6E Wireless Link
High Jitter (P₉₉ > 450 ms)

Non-blocking Async Leases

AXI / PCIe Gen3 Crossbar
4–8 GB/s Shared UMA Bus

QoS: Video DMA Throttled

Dedicated SRAM / DMA Mailbox
Zero-Copy Tensor Buffers

Burst Weight Streaming

Shared SRAM / SPI / CAN-FD
Priority 0 (Zero Contention)

< 250 μs Emergency Reflex